Polarisation

Book 2 §20 units 17–22 (taught from Ghatak ch. 22, §24.5) · pattern of the Nov 2024 / Nov 2025 papers, doubled

HOW THIS PAPER WORKS

A real paper has 12 short, 7 paragraph and 2 essay questions. This paper has **twice as many** (24 + 14 + 4), all from Module 4. Together they test every concept of the module; no concept is the main point of two questions.

Time	4 hours (or two sittings of 2 hours)
Maximum	120 marks

PYQ N24 Q10 = asked in Nov 2024, question 10. If a concept was asked in several years, all of them are listed. "adapted" = a 1-mark fill-in or True/False turned into a short answer.

Section A · Short answer

Answer all questions in two or three sentences. Each carries 2 marks. Ceiling 40.

1. Define polarization.

PYQ N24 Q10

2. Distinguish between the plane of vibration and the plane of polarisation.

3. Why can light waves be polarised but sound waves in air cannot?

4. What is unpolarised light? How is it represented in diagrams?

5. State Malus' law.

6. Unpolarised light of intensity I_0 passes through two polaroids whose pass axes are at 60°. Find the emergent intensity.

7. What is a Polaroid? On what principle does it work?

8. Why is light scattered at 90° by air molecules plane polarised?

9. State Brewster's law and what is Brewster window?

   PYQ N20 Q17

10. Find the polarising angle for glass of refractive index 1.5 and the angle of refraction at that incidence.

11. "Both ordinary ray and extraordinary ray are plane polarised." True or false? Justify.

    PYQ N19 Q8 · adapted

12. What is the optic axis of a crystal? Distinguish uniaxial and biaxial crystals.

13. "In a negative uniaxial crystal, velocity of O-ray is less than velocity of E-ray." True or false? Explain.

    PYQ N22 Q9 · adapted

14. What is a principal section of a crystal? How do the O and E vibrations lie with respect to it?

15. Define Huygens' explanation of double refraction.

    PYQ N24 Q9

16. Light falls normally on a calcite plate whose optic axis is perpendicular to its faces. What do you observe? Why?

17. How is the ordinary ray eliminated in a Nicol prism? Find the critical angle at the balsam layer ($\mu_o = 1.658$, $\mu_{balsam} = 1.55$).

18. What do you mean by a half-wave plate?

    PYQ N20 Q16

19. What is a quarter-wave plate? Write the expression for its thickness.

20. Plane polarised light falls on a quarter-wave plate, and separately on a half-wave plate, with its vibration at 45° to the optic axis. What emerges in each case?

21. Compare a Nicol prism with a Polaroid as a polariser.

22. Distinguish between right and left circularly polarised light.

23. Discuss optical activity briefly.

PYQ N25 Q7

24. Define the term specific rotation.

PYQ N20 Q5

Section B · Paragraph / problem

Answer all questions in about half a page to one page. Each carries 5 marks. Ceiling 60.

25. Derive Malus' law. Unpolarised light of intensity I_0 falls on three polaroids; the first and last are crossed and the middle one is at 45° to both. Find the emergent intensity.

26. Derive Brewster's law, and show that at the polarising angle the reflected and refracted rays are perpendicular.

27. Explain the production of plane polarised light by a pile of plates. Why is the transmitted light, and not the reflected light, used?

28. Calculate the thickness of a half wave plate and a quarter wave plate of quartz for a wavelength of 5000 Å. Here μE is 1.553 and μO is 1.544.

PYQ N20 Q28 · also N19 Q30 · N22 Q30 · N25 Q17

29. Velocity of E-ray along the optic axis of a calcite crystal is 1.81×10^8 m s^{-1} and its velocity perpendicular to the optic axis is 2.02×10^8 m s^{-1}. Find the velocity of E-ray along an axis that is at 45° angle with the optic axis.

PYQ N25 Q16

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30. Using Huygens' construction, show that when the optic axis is perpendicular to the plane of incidence, the E-ray obeys Snell's law with refractive index μ_e.

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31. Describe the construction and working of a Nicol prism.

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32. A wave plate is placed between crossed polarisers with its optic axis at angle φ to the first polariser. Show that $I = I_1 \sin^2 2\phi \, \sin^2(\delta/2)$. Find the intensity (in terms of the incident unpolarised I_0) for a QWP and for a HWP with φ = 45°.

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33. Left circularly polarised light ($\lambda = 5893$ Å) falls normally on a calcite plate of thickness 0.005141 mm cut with its optic axis parallel to its faces ($\mu_o - \mu_e = 0.17195$). Find the phase difference introduced and state the nature of the emergent light.

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34. Show that plane polarised light can be regarded as the sum of right and left circularly polarised light, and hence explain Fresnel's theory of optical rotation. Find the rotation produced by 1 mm of quartz along its optic axis if $|n_L - n_R| = 7 \times 10^{-5}$ and $\lambda = 6000$ Å.

35. Explain the method of producing plane, circularly and elliptically polarized light.

PYQ N20 Q23

36. Define double refraction. Explain Laurent's half-shade polarimeter.

PYQ N24 Q18

37. A calcite quarter-wave plate is made for λ = 4046 Å ($\mu_o - \mu_e = 0.18440$). Find its thickness. If it is used with λ = 7065 Å ($\mu_o - \mu_e = 0.16848$) and plane polarised light at 45° to its axis, what light emerges?

38. What is partially polarised light? When an analyser is rotated in a beam, the maximum intensity is three times the minimum. Find the degree of polarisation and describe the beam.

Section C · Essay

Answer any two, in about two pages each. Each carries 10 marks.

39. Describe how you produce circularly and elliptically polarized lights? How will you analyse the type of polarization?

PYQ N19 Q34 also N20 Q34 (last part)

40. Distinguish between (1) plane (2) circular and (3) elliptical polarization. Discuss the production of linearly polarized light by reflection and refraction.

PYQ N20 Q34 (first two parts)

41. Explain double refraction on the basis of Huygens' theory. Draw the wave surfaces for negative and positive uniaxial crystals, and use Huygens' construction for normal incidence when the optic axis is (a) parallel to the surface, (b) perpendicular to the surface, (c) inclined to the surface.

42. Obtain the resultant of two perpendicular simple harmonic vibrations of the same frequency with a phase difference δ. Discuss the cases $\delta = 0, \pi/2, \pi$ and general δ, and explain how these give plane, circular and elliptical polarisation.

Coverage check

TOPIC	QUESTIONS
T1 Polarisation, kinds	1, 2, 3, 4, 40
T2 Malus, partial polarisation	5, 6, 25, 38
T3 Polaroid, wire grid, scattering	7, 8
T4 Brewster, pile of plates, window	9, 10, 26, 27
T5 Double refraction, optic axis, crystals, v_e	11, 12, 13, 14, 29
T6 Huygens' explanation	15, 16, 30, 41
T7 Nicol prism	17, 21, 31
T8 Perpendicular vibrations, handedness	22, 42
T9 Wave plates	18, 19, 20, 28, 33, 37
T10 Production; plate between polarisers	35, 32, 39
T11 Analysis	39, 40
T12 Optical activity, specific rotation, Laurent	23, 24, 34, 36

All 18 Module 4 past questions (2019–2025) are included, either directly or as an "also asked" tag.